

VANGUARD AEROSPACE: RMD-04

ATMOSPHERIC & TRANS-MEDIUM

MANEUVERS

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DIRECTOR'S NOTE: *The Omni-Drift Calibration sequence in §3 has been completely rewritten following the 2026-05-01 incident where Pilot Chen's GWA pulse timing was off by 0.008 seconds, resulting in an uncontrolled tumble at Mach 12. The new 8-step calibration sequence includes mandatory cross-checks with the Power Systems team per RMD-01. All pilots must re-qualify on the updated procedure. — Col. Webb*

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document contains maneuver procedures that, if disclosed, could enable adversary reverse-engineering of TARTARUS-1 flight characteristics.

1. STANDARD OPERATING PROCEDURES (SOP)

The TARTARUS-1 craft exhibits flight characteristics that violate Newtonian physics as traditionally understood. Pilots must unlearn traditional aerodynamics. The craft does not generate lift via airfoil interaction; it generates thrust via graviton wavefront projection (RMD-01 §1.3) and maintains stability via the Inertial Nullifier (RMD-03 §2). Conventional control surfaces (ailerons, elevators, rudders) are present on the hull but are decorative — they serve only to maintain the appearance of a conventional aircraft during low-altitude operations over populated areas.

1.1 The “Right-Angle” Turn

At velocities exceeding Mach 5, the craft can execute an instantaneous 90-degree change in vector. This maneuver is physically impossible for any conventional aircraft and produces radar signatures that appear as a “disappearing-reappearing” phenomenon.

Procedure:

Step	Action	Timing	RMD Reference
1	Verify MCF integrity >96%	Pre-maneuver	RMD-01 §1.1
2	Verify Nullifier calibrated to pilot’s current mass	Pre-maneuver	RMD-03 §2.1
3	Verify Nullifier active and power draw <250 MW	Pre-maneuver	RMD-03 §2.2
4	Disengage aerodynamic control surfaces	T+0.000 s	Decorative at this speed
5	Pulse port-side GWA for exactly 0.04 seconds	T+0.000 s	RMD-01 §1.3
6	Re-engage forward GWA at full thrust	T+0.040 s	Must occur within 0.002 s of port pulse end
7	Verify new heading	T+0.050 s	QEA confirms via CSI intent vector (RMD-02 §3.1)
8	Re-engage aerodynamic control surfaces (if below Mach 3)	T+0.500 s	Only if speed has bled below Mach 3

Critical Timing: The port-side GWA pulse MUST be exactly **0.04 seconds** (± 0.001 s). - IF pulse duration < 0.039 s: Insufficient thrust for 90° turn; craft will execute a 45° deflection instead, resulting in off-heading condition. - IF pulse duration > 0.041 s: Excessive thrust; craft will execute a 135° turn, potentially entering a pursuit vector toward friendly assets.

Note: Radar systems will interpret this as the craft disappearing and reappearing. This is the basis for the majority of “UAP” (Unidentified Aerial Phenomena) reports in the vicinity of test operations.

1.2 The “Split-S” Reversal

At velocities exceeding Mach 10, the craft can reverse its direction of travel by 180 degrees in under 0.3 seconds. This maneuver produces extreme G-forces that would be fatal without the Inertial Nullifier (RMD-03 §2).

Procedure:

Step	Action	Timing	RMD Reference
1	Verify Nullifier power draw <210 MW	Pre-manuever	RMD-03 §2.2
2	Pilot visualizes “reverse heading” via CSI	Pre-manuever	RMD-03 §1
3	Disengage all aerodynamic surfaces	T+0.000 s	N/A
4	Pulse BOTH port and starboard GWA simultaneously for 0.06 s	T+0.000 s	RMD-01 §1.3
5	Engage forward GWA in reverse polarity	T+0.060 s	Must occur within 0.002 s of dual pulse end
6	Verify new heading (180° from original)	T+0.100 s	QEA confirms via CSI (RMD-02 §3.1)
7	Re-engage forward thrust at nominal	T+0.300 s	N/A

Maximum Safe Speed for Split-S: Mach 15. Above Mach 15, the graviton shear at the reversal point exceeds the Nullifier’s compensation capacity (RMD-03 §2.2), resulting in potential spinal compression injuries.

1.3 Vertical Zoom Climb

The craft can accelerate from level flight to vertical climb at **120 G** (compensated by the Inertial Nullifier). This maneuver is used to rapidly gain altitude or evade surface-to-air threats.

Step	Action	Timing	RMD Reference
1	Verify MCF >97%, Nullifier calibrated, GWA nominal	Pre-maneuver	RMD-01 §1.1, RMD-03 §2.1
2	Rotate nose to vertical (or pilot visualizes vertical via CSI)	T+0.000 s	RMD-03 §1
3	Engage forward GWA at 120% thrust	T+0.000 s	RMD-01 §1.3
4	Maintain for 2.0 seconds	T+0.000– 2.000 s	Reaches ~40,000 ft from sea level
5	Reduce GWA to nominal	T+2.000 s	N/A

Power Draw Warning: The Vertical Zoom Climb draws **1,100 MW** from the Casimir Array. Per RMD-01 §5, this exceeds the ECLS and Sensor shed thresholds. **IF** total Casimir output drops below 900 MW during the climb, **THEN** sensors and ECLS will be shed. Notify Astrometrics per RMD-02 before initiating this maneuver.

2. RADAR EVASION & METAMATERIALS

The hull is composed of a programmable metamaterial that absorbs and re-emits radar waves out of phase. The metamaterial consists of **12.7 million unit cells**, each individually addressable by the QEA. The system can dynamically adjust the hull’s electromagnetic response to any radar band in real-time.

2.1 Radar Cross-Section (RCS) Matrix — Standard Configuration

Radar Band	Frequency	Wavelength	Hull Response	Signature on Enemy Radar	RCS (m ²)	Effective Signature
X-Band	8–12 GHz	2.5–3.75 cm	Phase Cancellation	Flock of birds / Weather anomaly	0.001	0.001 m ² (insect-level)
S-Band	2–4 GHz	7.5–15 cm	Frequency Shift	Commercial airliner (Boeing 737)	100.0	100.0 m ² (spoofed)
L-Band	1–2 GHz	15–30 cm	Total Absorption	Stealth (No return)	< 0.0001	Below noise floor
C-Band	4–8 GHz	3.75–7.5 cm	Phase Cancellation	Weather balloon / Small drone	0.01	0.01 m ²
UHF	300 MHz–1 GHz	0.3–1 m	Partial absorption	Small general aviation aircraft	1.0	1.0 m ²
VHF	30–300 MHz	1–10 m	Active scatter redistribution	Ground clutter / Terrain return	0.1	0.1 m ²

2.2 Radar Cross-Section Matrix — Atmospheric Density vs. Radar Detectability

The RCS of the TARTARUS-1 craft varies with atmospheric density, as the metamaterial's phase-cancellation effectiveness depends on the ambient medium's electromagnetic properties.

Altitude	Atmospheric Density (kg/m ³)	X-Band RCS (m ²)	S-Band RCS (m ²)	L-Band RCS (m ²)	Overall Detectability
Sea Level	1.225	0.001	100.0 (spoofed)	< 0.0001	LOW — spoofs as commercial aircraft
1,000 m	1.112	0.001	100.0	< 0.0001	LOW — same spoofing profile
5,000 m	0.736	0.002	85.0	< 0.0001	LOW-MODERATE — spoofed signature slightly degraded
10,000 m	0.414	0.005	60.0	< 0.0001	MODERATE — spoofed signature reduced
20,000 m	0.089	0.015	30.0	0.001	MODERATE — partial L-band detection possible
30,000 m	0.018	0.040	15.0	0.005	MODERATE-HIGH — multiple bands may detect
50,000 m	0.001	0.100	5.0	0.010	HIGH — reduced metamaterial effectiveness
100,000 m	0.000001	0.500	1.0	0.050	VERY HIGH — near-vacuum; metamaterial degraded
Vacuum (Space)	0.000000	1.000	0.500	0.100	MAXIMUM — rely on fold evasion (RMD-02 §1)

Critical Note: At altitudes above 50,000 m, the metamaterial’s phase-cancellation effectiveness degrades because the ambient medium lacks sufficient molecular density to support the interference pattern. At these altitudes, the primary evasion method shifts from metamaterial cloaking to spatial fold evasion per RMD-02 §1. **IF** operating above 50,000 m, **THEN** notify Astrometrics per RMD-02 that fold evasion is the primary radar countermeasure.

2.3 Active Radar Countermeasures

In addition to the metamaterial hull, the TARTARUS-1 carries an **Active Radar Countermeasure (ARC)** suite that can project false radar returns at user-specified distances and headings.

ARC Mode	Function	False Return Range	False Return Heading	RMD Reference
Ghost	Projects single false target at random heading	5–50 km	Random	N/A
Swarm	Projects multiple false targets mimicking formation flight	2–20 km	User-defined	N/A
Mirage	Projects false terrain return to mask altitude	0–5 km (ground)	Below craft	N/A
Blackout	Total radar emission suppression (passive mode)	N/A	N/A	N/A
Echo	Mirrors real radar returns with 0.5 s delay (confuses tracking)	Current position +0.5 s	Same heading	N/A

3. HOVER AND “OMNI-DIRECTIONAL” DRIFT

The craft can remain perfectly stationary in the air while rotating on any axis. To achieve Omni-Drift, the pilot must visualize a “sphere of stillness” while mentally rotating the horizon line via the CSI (See RMD-03 §1).

3.1 Omni-Drift Calibration Sequence

The Omni-Drift mode requires precise GWA pulse timing to maintain stationary hover while allowing multi-axis rotation. The following 8-step calibration sequence must be completed before any Omni-Drift operation.

Step	Action	GWA Pulse Timing	Verification	RMD Reference
1	Verify MCF >98%, Nullifier calibrated, CSI coherence >98%	N/A (pre-calibration)	All three systems nominal	RMD-01 §1.1, RMD-03 §1.1, RMD-03 §2.1
2	Engage all 4 GWA projectors at hover thrust (15% nominal)	Continuous, symmetric	Craft maintains altitude ± 2 m	RMD-01 §1.3
3	QEA calculates Omni-Drift tensor	N/A	Tensor convergence in <0.003 s	RMD-02 §1.2
4	Initiate GWA pulse sequence Alpha : Port at 12.4 Hz, Starboard at 12.4 Hz, Aft at 12.4 Hz, Forward at 12.4 Hz	12.4 Hz (80.6 ms period), 0.04 s pulse width, 0.0406 s off-time	All 4 projectors synchronized to within ± 0.0001 s	RMD-01 §1.3
5	Verify hover stability via CSI: Pilot visualizes "sphere of stillness"	N/A	QEA confirms FSI >0.95 (RMD-02 §1.2)	RMD-03 §1
6	Initiate GWA pulse sequence Beta : Adjust pulse width to 0.02 s on alternating axes	0.02 s pulse width (alternating port/starboard and fore/aft)	Craft rotates on single axis while maintaining hover	RMD-04 §3.1
7	Pilot mentally rotates horizon line via CSI to direct drift axis	N/A	QEA tracks intent vector (RMD-02 §3.1)	RMD-03 §1
8	Cross-check: Verify all GWA power draws <150 MW each, total GWA power <600 MW	N/A	Power Systems confirm	RMD-01 §5

Critical Timing Notes: - The 12.4 Hz pulse frequency is critical. This frequency produces a graviton wave pattern that "balances" the craft in a standing-wave configuration. - **IF** pulse frequency drifts to <10.0 Hz: The standing wave destabilizes; craft will oscillate vertically (± 5 m). - **IF** pulse frequency drifts to >15.0 Hz: The standing wave compresses; craft will experience localized time dilation effects (see RMD-05 §1). - **IF** pulse width deviates by >0.005 s from specified values: Asymmetric thrust will cause uncontrolled rotation.

3.2 Omni-Drift Operational Limits

Parameter	Limit	Rationale	RMD Reference
Maximum Altitude	30,000 m	Metamaterial degradation above this (§2.2)	RMD-04 §2.2
Maximum Duration	4 hours	GWA thermal limits; cooldown required	RMD-01 §6
Maximum Occupants	2	Nullifier field coherence (RMD-03 §2.2)	RMD-03 §2.2
Maximum Rotation Rate	360°/s	Above this, CSI coherence degrades due to vestibular conflict	RMD-03 §1.1
Maximum Drift Velocity	50 knots	Beyond this, Omni-Drift transitions to standard flight	RMD-04 §1

3.3 Hover-to-Flight Transition

Step	Action	RMD Reference
1	Pilot visualizes “expansion of sphere” via CSI	RMD-03 §1
2	QEA shifts GWA pulse sequence from Alpha to standard forward thrust	RMD-02 §3.1
3	Increase GWA thrust from hover (15%) to cruise (60%) over 2.0 seconds	RMD-01 §1.3
4	Disengage Omni-Drift tensor calculation	RMD-02 §1.2
5	Verify forward velocity >20 knots (Omni-Drift disengaged)	N/A
6	Engage aerodynamic control surfaces if below Mach 0.5	N/A

4. FORMATION FLIGHT PROTOCOLS

TARTARUS-1 craft can operate in formation, but the graviton wavefronts from multiple craft can interfere destructively. The following protocols prevent wavefront interference.

Formation Type	Minimum Separation	Maximum Craft	RMD Reference
Line Abreast	500 m	4	RMD-01 §1.3
Trail	1,000 m	6	RMD-01 §1.3
Echelon	750 m	4	RMD-01 §1.3
Diamond	600 m	4	RMD-01 §1.3
Omni-Drift (stationary)	2,000 m	2	RMD-04 §3

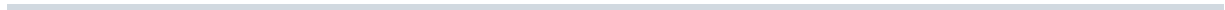
IF two craft come within minimum separation, **THEN** the graviton wavefronts will constructively interfere, producing a localized gravitational anomaly that can pull the craft together. **IF** constructive interference exceeds 6.0 TeV at any point between craft, **THEN** both craft must immediately execute emergency separation (port GWA pulse for craft 1, starboard GWA pulse for craft 2, 0.04 s each per §1.1).

5. ADVERSARY ENGAGEMENT PROCEDURES

CLASSIFICATION NOTE: This section is compartmented under **AZTHREUS-TACTICAL**. Access requires additional need-to-know authorization from the Base Commander.

5.1 Threat Response Matrix

Threat Type	Detection Range	Response	GWA Configuration	RMD Reference
Surface-to-Air Missile (SAM)	50 km (radar lock)	Omni-Drift evasion (§3) + Fold to safe coordinates	Hover thrust, no forward	RMD-02 §1, RMD-04 §3
Air-to-Air Missile (AAM)	20 km (radar lock)	Right-Angle Turn (§1.1) + ARC Swarm mode (§2.3)	Forward thrust, port pulse 0.04 s	RMD-04 §1.1, §2.3
Ground-Based Radar	200 km (detection)	Metamaterial cloaking (§2.1) + ARC Ghost mode	Standard cruise	RMD-04 §2.1, §2.3
Airborne Early Warning (AEW)	400 km (detection)	Fold evasion at Class I or higher	Forward thrust, prepare fold	RMD-02 §1
Directed Energy Weapon (DEW)	10 km (effective range)	Immediate fold evacuation	Full GWA thrust	RMD-02 §1
Nuclear Detonation	Unlimited	Fold to maximum safe distance; initiate Tachyon Shielding	Full GWA thrust	RMD-05 §1



6. CROSS-REFERENCE INDEX

Reference	Document	Section	Topic
RMD-01 §1.1	Power & Propulsion Interface	Element 115 Decay Chamber	MCF requirements for maneuvers
RMD-01 §1.3	Power & Propulsion Interface	Gravity Wave Amplifiers	GWA pulse timing and thrust specifications
RMD-01 §3	Power & Propulsion Interface	Emergency Shutdown	COLD IRON during flight emergencies
RMD-01 §5	Power & Propulsion Interface	Power Distribution	Power shedding during high-G maneuvers
RMD-01 §6	Power & Propulsion Interface	Maintenance Schedules	GWA inspection requirements
RMD-02 §1	Spatial Manifold Navigation	Non-Euclidean Routing	Fold evasion procedures
RMD-02 §1.2	Spatial Manifold Navigation	Alcubierre-White Tensor	Fold tensor calculations for evasion
RMD-02 §3.1	Spatial Manifold Navigation	CSI-QEA Integration	Intent vector for drift control
RMD-03 §1	Neural Sync & Inertial Damping	Cortical Shunt Interface	CSI coherence for maneuver visualization
RMD-03 §1.1	Neural Sync & Inertial Damping	Sync Coherence Requirements	Coherence limits per maneuver type
RMD-03 §2	Neural Sync & Inertial Damping	Inertial Nullifiers	G-force compensation during maneuvers
RMD-03 §2.1	Neural Sync & Inertial Damping	Nullifier Calibration	Mass calibration for drift operations
RMD-05 §1	Emergency Containment & Scuttle	Tachyon Radiation	Tachyon effects from high-frequency GWA pulses
RMD-05 §3	Emergency Containment & Scuttle	Protocol Omega	Last-resort scuttle during adversary engagement

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